

Sensing Presence: Modulating Social Robot Behavior Through Proximity Detection

Elizabeth Sánchez R.¹, María Gaitán-Padilla¹ *IEEE Student Member*, Vinicius Mageste Coimbra¹, João Pedro Sandrini Milanezi¹, Maria José Pontes¹, Ricardo Mello¹ *IEEE Member*, Marcela Múnera², Carlos A. Cifuentes² *IEEE Senior Member* and Camilo A. R. Diaz¹ *IEEE Member*

Abstract—This study evaluates the integration of proximity sensing technologies into a Social Assistive Robot (SAR) to enhance interaction quality in therapeutic contexts. Two sensor types, a laser-based Time-of-Flight sensor and a radiofrequency-based sensor were compared in terms of detection range, accuracy, precision, robustness, and ease of integration. The laser-based sensor demonstrated superior performance with a score of 91.03% and was embedded into the SAR. A perception protocol was developed to activate simple social behavior based on user distance, and the interaction was assessed with 10 neurotypical adults under autonomous and Wizard of Oz conditions. Results showed that both interaction modes were positively received, with no significant differences in user perception. These findings support the feasibility of incorporating presence-based functionality in SARs to facilitate more natural and engaging human-robot interaction, laying the groundwork for future use in interventions with children with Autism Spectrum Disorder (ASD).

Index Terms—Social Robot, Human-Robot Interaction, Proximity sensing.

I. INTRODUCTION

Social Assistive Robots (SARs) have gained significant attention in healthcare and therapeutic contexts due to their ability to engage users through natural and multimodal interactions [1]. In interventions with children diagnosed with Autism Spectrum Disorder (ASD), these robots have shown potential to support the development of social, communicative, and emotional skills [2]. Their predictable behavior, physical embodiment, and capacity for social engagement make them especially suitable for children with ASD, who often benefit from structured and consistent interactions [3], [4].

Detecting and responding to the presence of humans by proximity sensing is a crucial factor in making interactions with SARs meaningful, which serves as a primary mechanism to initiate social and context-aware behavior [1]. Mitsunaga *et al.* demonstrated that adapting robot behaviors based on human nonverbal cues can enhance user comfort and engagement, such as interaction distance, gaze, and motion speed, especially

when these adaptations are guided by subconscious feedback rather than explicit user input [5].

Beyond initiating interaction, proximity sensing serves as a complement to vision and touch in Human-Robot Interaction (HRI), particularly in dynamic or overloaded environments, while vision systems are effective for long-range perception, they often struggle with occlusions and variable lighting [6]. Proximity sensors help address some of these limitations by enabling continuous, contactless detection, allowing robots to anticipate human presence even when visual or tactile cues are unavailable. This capability enhances both safety and responsiveness, which is valuable in therapeutic contexts where user comfort and social appropriateness are essential [1], [5]. In addition, proximity detection supports the regulation of interpersonal distance, helping robots respect intimate (0–45 cm), personal (45–120 cm), and formal (120–365 cm) zones - a key to maintaining user comfort [7].

Two of the most relevant technologies for proximity sensing in HRI are laser-based (optical) and radiofrequency-based (radar) sensors, each with distinct advantages [6]. Laser-based sensors, including reflected light intensity (O-RLI), time-of-flight (O-ToF), and triangulation types are widely used for their high spatial resolution, compact size, and fast response times. These features make them ideal for tasks such as manipulation, object detection, and collision avoidance in short-range, structured environments [8], [9]. On the other hand, radiofrequency-based sensors, especially Frequency-Modulated Continuous Wave (FMCW) radars, provide reliable performance regardless of lighting conditions or surface properties. They can simultaneously measure distance and velocity, penetrate certain materials, and operate without a direct line of sight [10], [11].

This study aims to evaluate and implement proximity sensing technologies for SARs, with a focus on improving HRI in therapeutic contexts. We compared laser-based and radiofrequency-based sensors in terms of sensitivity, range, precision, integration, and robustness. The selected sensor was then integrated into a SAR and tested with healthy subjects to validate its effectiveness in detecting human presence and enabling socially appropriate responses, laying the foundation for future use in therapies of children with ASD.

II. METHODOLOGY

This section outlines the framework used to compare two proximity sensor technologies for integration into a social robot in therapy, covering performance analysis, sensor in-

This work is partially supported by FAPES (EXTRATO ATA DA 6ª REUNIAO CCAF/2022 – “CPID 2.2”, FAPES/ CNPq/Decit – PPSUS 17/2024, 2023–XDHHW), CNPq (306323/2024-9, 404111/2023-8, and 402739/2024-8), and IEEE RAS SPARX.

¹ E. Sánchez, M. Gaitán-Padilla, V. Coimbra, M. J. Pontes, R. Mello, and C. A. R. Diaz are with the Telecommunications Laboratory (LabTel), Electrical Engineering Department, Federal University of Espírito Santo, Vitória, Brazil. (elizabeth.rodriquez, maria.padilla, vinicius.coimbra, joao.milanezi)@edu.ufes.br, (maria.pontes, ricardo.c.mello, camilo.diaz)@ufes.br

² M. Munera and C. A. Cifuentes are with the Bristol Robotics Laboratory, University of the West of England, Bristol, UK. marcela.munera@uwe.ac.uk, carlos.cifuentes@uwe.ac.uk

tegration, participant description, experimental procedure, and evaluation methods.

A. Proximity sensors

To evaluate the most suitable proximity sensing, was conducted a comparative study of two distinct sensors: the VL53L3CX Time-of-Flight (ToF) (STMicroelectronics, Switzerland) laser-based sensor and the HLK-LD2420 24 GHz radiofrequency-based sensor (Hi-Link Electronic Co., Ltd, China). The VL53L3CX is a compact, Time-of-Flight (ToF) sensor. It operates using a 940 nm invisible class 1 laser emitter and a Single Photon Avalanche Diode (SPAD) array to measure the time it takes for emitted light to reflect off a target and return to the sensor. This method allows for absolute distance measurement, regardless of the target's color or reflectivity. Unlike conventional infrared sensors that estimate distance based on reflected signal intensity, this sensor uses the round-trip time of the laser pulse to calculate distance with high precision [12]. The sensor features a Field of View (FoV) of 25° and supports detection distances of up to 4 m [12]. The one-way distance d to the target is calculated as shown in Equation 1, where c is the speed of light and Δt is the round-trip ToF.

$$d = \frac{c\Delta t}{2} \quad (1)$$

The HLK-LD2420 is a 24 GHz radar sensor developed for accurate human presence detection. It operates using Frequency-Modulated Continuous Wave (FMCW) technology, which emits continuous radio waves with linearly modulated frequency and measures the frequency difference between the transmitted and received signals. This corresponds to the time delay caused by the signal reflecting off a target and returning to the sensor [13]. This technique allows the sensor to detect motion, micro-motion, and stationary human presence. The radar includes one transmitter and one receiver antenna. The distance to a target is estimated using the FMCW radar ranging that is given in Equation 2 where c is the speed of light, Δf is the beat frequency (Hz), B is the modulation bandwidth (Hz) and T is the sweep time or modulation period (s).

$$d = \frac{c\Delta f}{2BT} \quad (2)$$

B. Robot implementation

1) *Robotic Platform:* The CASTOR robot is a cost-effective, open-source platform developed to support physical HRI in therapeutic environments for children with ASD [14]. Its structural design prioritizes safety, incorporating actuation mechanisms such as elastic joints and biologically inspired tendons. The robot features 14 degrees of freedom, powered by seven Dynamixel MX/AX motors (Robotis, France) and five MG996 servomotors (TowerPro, Taiwan), complemented by a passive damping system to absorb impact. In terms of interactive capabilities, CASTOR enables arm and neck movements, simulated emotional expressions, and a huggable system to encourage physical engagement. Its 3D-printed

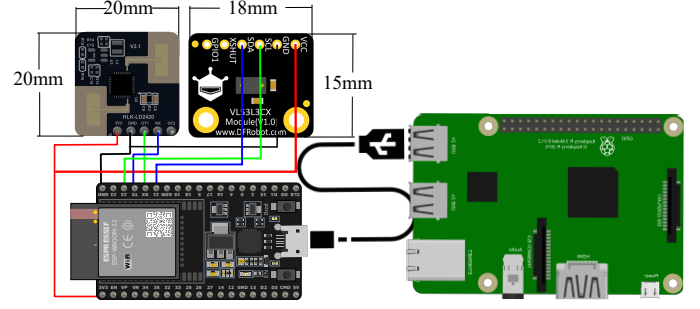


Fig. 1: Experimental setup showing the connection of the RF (24 GHz, 20×20 mm) and the laser-based sensor (15×18 mm) to an SBC.

modular architecture ensures ease of assembly, maintenance, and customization.

2) *Sensor Integration:* Following the comparative evaluation of the proximity sensors, the selected one was integrated into the CASTOR robot to enable real-time detection of human distance. As illustrated in Figure 1, both sensors were initially connected to a microcontroller ESP 32 (Espressif Systems, China) and a single board computer (SBC) using serial communication to facilitate testing and data acquisition.

To integrate the sensor into the robot's structure, a 3D-printed piece was developed, designed to fit the dimensions of the selected sensor. This component align the sensor's FoV with the intended interaction area of the robot. The sensor's data processing scripts were added to the CASTOR control architecture. The sensor's output was linked to the robot's behavior controller, allowing the system to create audio responses and antenna colors through a Light Emitting Diode (LED) based on proximity, following the perception protocol.

C. Participants

The evaluation of the perception protocol was conducted with 10 neurotypical adults aged between 20 and 30 years, all in healthy conditions. This work was approved by the ethics committee of the Federal University of Espirito Santo (85461424.7.0000.5542).

D. Procedure

1) *Comparative technologies:* The two proximity sensors were tested under the same conditions. Distances were marked from 0.2 to 1.30 meters, and human approach trials were conducted in increments of 10 centimeters every 10 seconds. For each sensor, detection readings were collected at angular positions matching their FoV: −12.5° to 12.5° for the laser-based (25° FoV) and −60° to 60° for the radiofrequency-based (120° FoV). In addition to range tests, precision was measured at each fixed distance by repeating presence and absence cycles. This helped calculate accuracy, precision, relative error rate, and precision rate. To see how each sensor worked in different lighting, the relative error rate was measured under normal, low, and high light levels. Light intensity was measured to keep conditions consistent. At each light level, tests were repeated from 0.2 to 1.30 meters.

2) *Perception protocol*: The perception protocol was designed to activate simple social behavior from the robot based on proximity detection, using visual and auditory cues. It was structured into three proximity zones commonly detected in social interactions: intimate (≤ 45 cm), personal (45 - 120 cm), and formal (≥ 120 cm) [7]. Depending on the zone, the robot activated specific LED color cues and audio messages:

- Intimate: The robot turned on a green LED and initiated interaction with a greeting and follow-up questions. It continues with “Simon says” game, and finally says “Good bye”.

- Personal: The robot displayed a yellow LED and encouraged the person to come closer with phrases like “Come closer”. If the person remained at this distance after two attempts, it proceeded to say “Good bye”.

- Formal: A blue LED indicated the person stayed far. If the session was about to start, the robot invited the participant to approach. If it was the end, it concluded the interaction with “See you next time!”.

To assess how the proximity system affects user perception, participants were split into two groups. One group used the robot with the proximity feature on, while the other did the same activity without it. In the second session, they switched conditions. When proximity sensing was off, the robot followed the activity using a Wizard of Oz (WoZ) approach, ignoring the user’s distance.

E. Evaluation variables

1) *Comparative technologies*: Each sensor was evaluated using a set of quantitative and qualitative metrics. The following variables were considered:

- Detection Range (DR): It was the distance at which the sensor reliably detected human distance across three angular zones: the central axis (0°) and the minimum and maximum field-of-view angles of each sensor in 10 cm increments. To quantify performance, the range was divided into the three proximity zones, each zone was evaluated based on the measurements with a relative error below 20%. If at least 80% of the values were below threshold, a point was assigned per valid zone, with 3 indicating detection across the entire range.

- Relative error rate (E%): This metric quantifies how far a sensor’s output deviates from the actual distance value during presence detection. It is calculated as presented in [15].

- Precision rate (P%): To assess the consistency of sensor measurements at a fixed distance was used this standard metric as described in [16].

- Precision and Accuracy - Classification-based (Pre CB and Acc CB): To evaluate detection performance in binary terms (presence vs absence), two metrics were calculated: precision and accuracy. The definitions of these metrics follow standard formulations used in classification tasks, as described in [17].

- Robustness to environmental conditions (RE): This was assessed by measuring the relative error rate at three controlled lighting conditions: low (RE LI), normal (RE NI), and high luminosity (RE HI). Ambient light levels were measured using a lux meter (DT-8820, CEM, China).

- Ease of Integration (EI): This qualitative metric reflects how easily each sensor could be embedded in the SAR, considering three factors: Mechanical compatibility (MC), Software compatibility (SC) and Hardware complexity (HC). The score was calculated using the weighted Equation 3.

$$EI = 0.4MC + 0.35SC + 0.25HC \quad (3)$$

Each component was rated on a likert scale from one to five by the investigators, where higher values indicate better compatibility or lower complexity and lower values indicate incompatibility and higher complexity.

- Sensor Selection Score (SS): To determine the most suitable sensor for this application, the Equation 4 was used. The evaluation variables were weighted according to their relevance to this social HRI. The variables are: E%, P%, EI, Acc CB, Pre CB, Rmax, RE LI, and RE HI.

$$SS = 0.025((1 - E\%) + P\% + (1 - RELI)) + 0.2(EI + AccCB + PreCB) + 0.1625(Rmax + (1 - REHI)) \quad (4)$$

2) *Perception protocol*: To evaluate the acceptance of the robot’s proximity-based interaction protocol, the Unified Theory of Acceptance and Use of Technology (UTAUT) framework combined with perception metrics was applied [14]. This perception questionnaire uses 13 metrics (Friendliness (F), Safety (S), Social Intelligence (SInt), Enjoyment (E), Naturalness (N), Social Influence (SI), Social Presence (SP), Perceived Ease of Use (PEOU), Attitude Toward Technology (ATT), Robot Safety (RS), Trust (PT), Perceived Utility (PU), and Anxiety (ANX)). The participants had the interaction randomly with the autonomous-based protocol and with the WoZ. After each interaction participants completed the questionnaire to capture their perception of the system’s usability and effectiveness, comparing both interactions. The survey also includes two open questions describing the robot and the interaction with three adjectives [18].

TABLE I: Comparative Analysis of Evaluation Variables.

Variables (%)	RF Sensor	Laser Sensor
DR	05.55	84.72
E	16.54	06.29
P	49.32	65.03
Pre	100.0	100.0
Acc	94.94	98.19
RE LI (97.20 lux)	19.39	08.07
RE HI (339.2 lux)	20.56	11.03
RE NI (237.6 lux)	16.54	06.29
EI	72.00	84.50
SS	72.53	91.03

III. RESULTS

The comparative analysis of the proximity sensors, summarized in Table I, highlights the performance of the laser-based sensor across most evaluation variables. It achieved a broader DR (84.72%) compared to the RF sensor (5.55%),

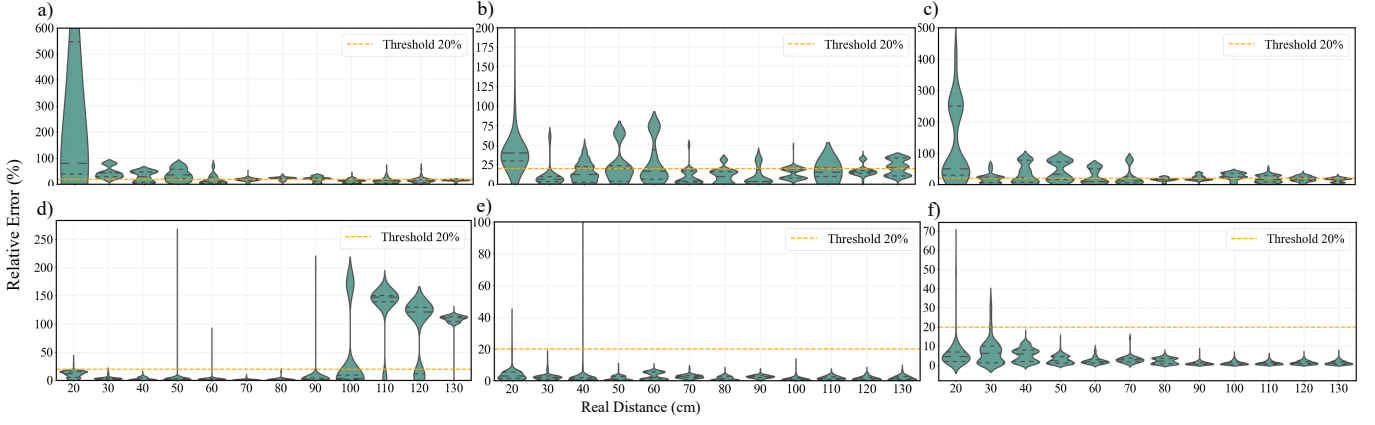


Fig. 2: Evaluation of detection range across distances for the sensors. The yellow dashed line represents the 20% threshold. Graphs show performance at different Fields of View (FoV): for the RF sensor: (a) Left-side: -60° FoV, (b) Center: 0° FoV, (c) Right-side: 60° FoV; and for the laser-based sensor: (d) Left-side: -12.5° FoV, (e) Center: 0° FoV, (f) Right-side: 12.5° FoV.

and a notably lower E% (6.29% vs. 16.54%). The laser sensor also demonstrated higher P% (65.03%) and Acc CB (98.19%) while maintaining 100% in Pre CB, as well as the RF sensor.

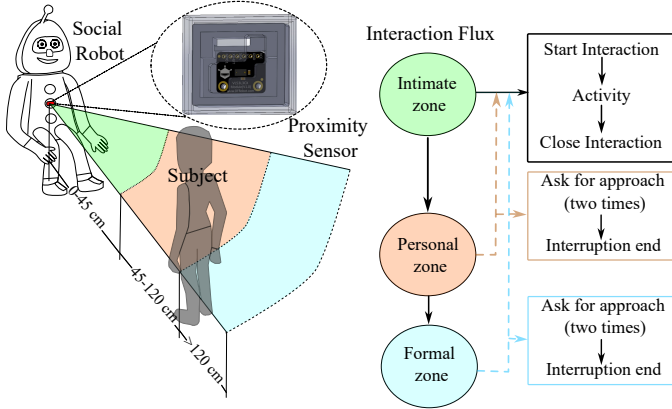


Fig. 3: Integration of the sensor into the SAR, showing the piece and the defined interaction zones. Also illustrates the interaction flux.

These results are supported by Figure 2, which presents the E% distributions across real distances and FoV for each sensor. The RF (Figure 2a–c) exhibited greater variability, with many distance points, especially at the extremes of its FoV, showing high dispersion and E% values well above the 20% threshold. In contrast, the laser-based sensor (Figure 2d–f) maintained a tight distribution of E% mostly below the threshold. Under varying lighting conditions, the laser sensor proved more robust, with RE below 12%, while the RF sensor exhibited higher RE rates (16–21%). Moreover, the laser sensor scored higher in terms of EI, with 84.5%, contributing to its overall SS of 91.03%, surpassing the RF sensor's 72.53%.

Figure 3 presents the integration of the proximity sensor into the SAR, emphasizing the custom-designed piece that securely the sensor and aligns its FoV with the user's frontal area. This enabled a reliable and consistent detection of the

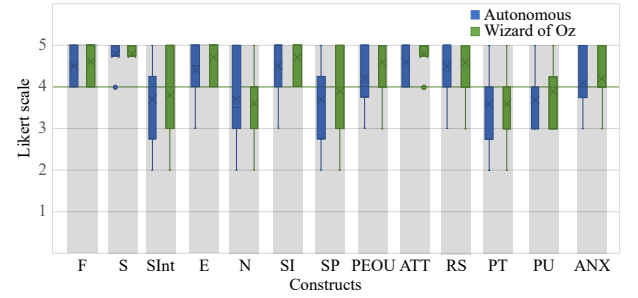


Fig. 4: Survey outcomes of perception of F, S, SInt, E, N, SI, SP, PEOU, ATT, RS, PT, PU, ANX comparing the proximity-based autonomous and the WoZ interaction.

user's distance in real time. The figure also defines the three operational zones: intimate, personal, and formal, which were essential for adapting the robot's behavior. Lastly, Figure 4 presents the comparison of user perception scores between the autonomous interaction and the WoZ condition across the 13 metrics evaluated. In general, the WoZ condition received slightly higher median scores than the autonomous condition across most metrics, without expressing real differences between both interactions in any metric. Both conditions showed similar median scores in constructs such as PEOU, SP, and S, with overlapping interquartile ranges indicating comparable perception. The boxplots also reveal slightly higher variability in user responses under the autonomous condition in the naturalness and ATT constructs.

IV. DISCUSSION

The RF sensor offers a low cost (approximately 2 USD), compact design, and the advantage of non-line-of-sight operation, making it an option for deployment in IoT and robotics applications [19]. However, while RF sensors can detect the presence of an object within a given range (Acc and Pre CB $\geq 94\%$), they do not provide precise distance measurements [20]. This limitation is consistent with the findings of this study,

where the RF sensor achieved a P% of 49.32% and an E% of 16.54%, indicating limited reliability for accurate proximity-based interaction. In contrast, the laser-based sensor offers greater functional flexibility, is lightweight, highly accurate, and demonstrates strong reliability [21]. These advantages are consistent with the results obtained in this study, where the laser sensor achieved a DR of 84.72%, a low E% of 6.29%, and a high P% of 98.19%. Additionally, laser-based systems can provide wider bandwidth, lower power consumption, and an enhanced potential for covert data transmission compared to radio-based alternatives [22].

The results of the user interactions did not reveal significant differences in the perception outcomes between the two trials, with generally positive evaluations across all metrics. This suggests that the robot's behavior, when controlled via the proximity sensor, was perceived as equally natural as when operated by an external technician. These findings support the integration of presence-based functionalities into the CASTOR robot without compromising interaction quality. However, lower scores were observed in the metrics related to SInt, N, SP, and PT, highlighting areas for improvement in the robot's ability to support socially engaging interactions, particularly during brief exchanges with adults. It is important to note that CASTOR was designed primarily for interactions with children, and future evaluations with the intended target population may yield higher perception ratings. Moreover, future work will build on this study by expanding the sample size and including participants from the intended target population. In addition, upcoming iterations will explore more flexible interaction protocols and evaluate the system in real-world therapeutic settings to better assess its applicability and adaptability.

V. CONCLUSIONS AND FUTURE WORK

The comparative evaluation of the laser-based and RF proximity sensors demonstrated the laser sensor's superior performance for integration into SARs. According to the defined evaluation criteria, the laser sensor obtained a higher SS of 91.03%, compared to 72.53% for the RF sensor, justifying its selection for integration into the robotic platform.

Moreover, the integration of proximity-based functionalities into the CASTOR robot demonstrated promising results, enabling socially appropriate interactions that were perceived as effective and natural. Although both the autonomous and WoZ interactions were well received, the findings support the feasibility of autonomous presence detection for social engagement. However, the study highlights the need to improve aspects as SInt, N and PT. Future work should focus on refining adaptive behaviors, validating the system with the focus population, and implementing the protocol in real therapeutic settings.

REFERENCES

- [1] C. Breazeal, K. Dautenhahn, and T. Kanda, "Social robotics," in *Springer Handbook of Robotics*, B. Siciliano and O. Khatib, Eds. Cham: Springer International Publishing, 2016, pp. 1935–1972.
- [2] C. A. Cifuentes, M. J. Pinto, and *et al.*, "Social robots in therapy and care," *Current Robotics Reports*, vol. 1, pp. 59–74, 2020.

- [3] M. J. Matarić and B. Scassellati, "Socially assistive robotics," in *Springer Handbook of Robotics*, B. Siciliano and O. Khatib, Eds. Cham: Springer International Publishing, 2016, pp. 1973–1994.
- [4] C. Liu, K. Conn, N. Sarkar, and W. Stone, "Online affect detection and robot behavior adaptation for intervention of children with autism," *IEEE transactions on robotics*, vol. 24, no. 4, pp. 883–896, 2008.
- [5] N. Mitsunaga, C. Smith, T. Kanda, H. Ishiguro, and N. Hagita, "Adapting robot behavior for human-robot interaction," *IEEE Transactions on Robotics*, vol. 24, no. 4, pp. 911–916, 2008.
- [6] S. E. Navarro, S. Mühlbacher-Karrer, H. Alagi, *et al.*, "Proximity perception in human-centered robotics: A survey on sensing systems and applications," *IEEE Transactions on Robotics*, 2021.
- [7] L. O. Kroczeck, M. Pfaller, B. Lange, M. Müller, and A. Mühlberger, "Interpersonal distance during real-time social interaction: Insights from subjective experience, behavior, and physiology," *Frontiers in psychiatry*, vol. 11, p. 561, 2020.
- [8] K. Hsiao, P. Nangeroni, M. Huber, A. Saxena, and A. Y. Ng, "Reactive grasping using optical proximity sensors," in *2009 IEEE International Conference on Robotics and Automation*, IEEE, 2009, pp. 2098–2105.
- [9] K. Koyama, M. Shimojo, T. Senoo, and M. Ishikawa, "High-speed high-precision proximity sensor for detection of tilt, distance, and contact," *IEEE Robotics and Automation Letters*, pp. 3224–3231, 2018.
- [10] M. Gerstmaier, A. Melzer, A. Onic, and M. Huemer, "On the safe road toward autonomous driving: Phase noise monitoring in radar sensors for functional safety compliance," *IEEE Signal Processing Magazine*, vol. 36, no. 5, pp. 60–70, 2019.
- [11] C. Stetco, B. Ubezio, S. Mühlbacher-Karrer, and H. Zangl, "Radar sensors in collaborative robotics: Fast simulation and experimental validation," in *2020 IEEE International Conference on Robotics and Automation (ICRA)*, IEEE, 2020, pp. 10 452–10 458.
- [12] STMicroelectronics, *VL53L3CX Time-of-Flight Ranging Sensor with Multi-Target Detection*, 2024.
- [13] Hi-Link Electronic Co., Ltd., *HLK-LD2420: 24GHz Human Presence Detection Sensor User Manual V1.1*, 2022.
- [14] M. Gaitán-Padilla, J. C. Maldonado-Mejía, L. Fonseca, *et al.*, "Physical human-robot interaction through hugs with castor robot," in *Social Robotics*, H. Li, S. S. Ge, Y. Wu, *et al.*, Eds., Cham: Springer International Publishing, 2021, pp. 814–818, ISBN: 978-3-030-90525-5.
- [15] B. N. DATTA, "Chapter 3 - some fundamental tools and concepts from numerical linear algebra," in *Numerical Methods for Linear Control Systems*, B. N. DATTA, Ed., San Diego: Academic Press, 2004, pp. 33–78, ISBN: 978-0-12-203590-6.
- [16] P. Dowd, "Accuracy and precision," in *Encyclopedia of Mathematical Geosciences*, Springer, 2023, pp. 1–4.
- [17] M. Hossin and M. N. Sulaiman, "A review on evaluation metrics for data classification evaluations," *International journal of data mining & knowledge management process*, vol. 5, no. 2, p. 1, 2015.
- [18] Z. Janeczko and M. E. Foster, "A study on human interactions with robots based on their appearance and behaviour," in *Proceedings of the 4th Conference on Conversational User Interfaces*, 2022, pp. 1–6.
- [19] M. Bolic, M. Rostamian, and P. M. Djuric, "Proximity detection with rfid: A step toward the internet of things," *IEEE Pervasive Computing*, vol. 14, no. 2, pp. 70–76, 2015.
- [20] Z. Saleem, F. Gustafsson, E. Furey, M. McAfee, and S. Huq, "A review of external sensors for human detection in a human robot collaborative environment," *Journal of Intelligent Manufacturing*, vol. 36, no. 4, pp. 2255–2279, 2025.
- [21] A. Bechar, S. Y. Nof, and J. P. Wachs, "A review and framework of laser-based collaboration support," *Annual Reviews in Control*, 2015.
- [22] J. Sofka, V. V. Nikulin, V. A. Skormin, D. H. Hughes, and D. J. Legare, "Laser communication between mobile platforms," *IEEE transactions on Aerospace and Electronic Systems*, vol. 45, no. 1, pp. 336–346, 2009.